

SENG1050

Lab 11

In this lab, you will focus on completing Assessment 3. With the available time after, you may explore some examples on RSA encryption and steganography.

[Task 1](#) - [Task 2](#)

Assessment 3 (Important!)

Work on Assessment 3 (Web-Based Assignment 2) and get your milestones graded by this week. Your assessment is also due this Friday.

Task 1 - RSA Encryption (Optional - Not in Exams)

RSA is a popular asymmetric encryption scheme. The scheme works by generating separate public and private keys that are mathematical inverses of each other for the purpose of encryption. This mathematical relationship allows a message encrypted with a public key to be decrypted with the matching private key and vice versa, a message encrypted with a private key can be decrypted with the matching public key.

The complete RSA scheme is presented in Lecture 11 and we will only reiterate the key generation process as follows:

1. Starting off, we choose two prime numbers, defined as p and q . In practice, these should be large prime numbers, but for simplicity, we will choose small numbers in this example, such that $p = 3$ and $q = 5$.
2. Both the public and private keys need a common element, n , such that $n = p * q$. In this example, $n = 15$.
3. Now, we need to calculate the modulus that we will be using, w , such that $w = (p-1) * (q-1)$. In this example, $w = 8$.
4. Now, we choose the unique element of the public key, e . e can be any number from 1 to w as long as it does not share any common factors with w (aside from 1).
In this example, we could choose 1, 3, 5, or 7 (but choosing 1 would result in no encryption being performed at all!). We could not choose a number like 6, because 2 is a factor of both 6 and 8.
Let us choose $e = 7$. The public key is $\langle e, n \rangle = \langle 7, 15 \rangle$.
5. Now, choose the unique element of the private key, d . d needs to be a multiplicative inverse of e under the modulus w , i.e., $e * d \bmod w = 1$.
As it happens, we can only have $d = 7$ as $e * d \bmod w = 7 * 7 \bmod 8 = 49 \bmod 8 = 1$.
The private key is $\langle d, n \rangle = \langle 7, 15 \rangle$, which coincidentally, is the same as the public key established in step 4.

The keys generated go against the principle of asymmetric key encryption and is a result of choosing small prime numbers in step 1. This further emphasizes the need of having large prime numbers, which will also be challenging for computers trying to perform prime factorization.

As a practice, consider the RSA public key $\langle e, n \rangle = \langle 23, 35 \rangle$. What would the private key be? (*Hint: Your first step would be determining the prime numbers, p and q , that were used.*)

Task 2 - Steganography (Optional - Not in Exams)

Cryptography (for confidentiality) encrypts a message so that even if it is intercepted, it cannot be read. Steganography hides the fact that the message exists at all. It is often used in publications or software to combat piracy. Let's have some fun with the following activity:

1. Download [steg.zip](#) available on Canvas and extract the contents to your preferred directory.
 2. Open [S-Tools.exe](#).
 3. Click and drag [zebras.bmp](#) into the S-Tools window. Expand the graphic to see the photo of zebras if necessary.
 4. Right click on the image and select Reveal. When prompted, enter the passphrase 'abc' (without the quotes). Leave the encryption algorithm as "IDEA" and click OK.
 5. Study the revealed files.
 6. Repeat steps 4 and 5 with [chi_in_bear.gif](#). Use the passphrase 'strange' (without the quotes) instead.
 7. Read the article [clair.pdf](#) to understand how steganography works.
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